

Genomic Data Equity: Addressing Ancestry Underrepresentation and Setting Equity Metrics for Polygenic Risk Score Deployment

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I. Introduction

The use of genomic data has become a fundamental pillar of modern biomedical research, from clinical decision-making and disease prediction to public health policy and even personalized medicine. Much of this work is driven by genome-wide association studies (GWAS) and polygenic risk scores (PRS), which use large-scale genomic datasets to model genetic contributions to disease risk. However, there is a fundamental flaw in the datasets used to elucidate these results: they remain overwhelmingly Eurocentric and, as such, are not representative of the entire population. According to Sirugo et al. (2019), over 78% of participants in GWAS Catalogs were of European ancestry, despite Europeans representing less than 16% of the entire global population. The consequence of this oversight is not just a demographic oversight; it fundamentally shapes the genetic architecture used to estimate risk, producing evidence that is not generalizable across populations.

Fundamental principles of population genetics amplify the consequences of this bias. While humans are genetically more similar than different overall, most differences relevant to GWAS and PRS arise from variation in haplotype frequency and linkage disequilibrium (LD) structure across global populations rather than from population-specific “race genes” (Rosenberg et al., 2002; 1000 Genomes Project Consortium, 2015). Populations of individuals with African ancestry exhibit shorter haplotype blocks and lower LD due to a more extended recombination history (International HapMap Consortium, 2007; Tishkoff et al., 2009). In contrast, populations that experienced bottlenecks during migration out of Africa, such as Europeans and East Asians, have longer LD blocks and reduced haplotype diversity (Schaffner et al., 2005; Gravel et al., 2011). Because GWAS relies on identifying tag SNPs that correlate with causal variants through LD, associations discovered in one ancestry often fail to replicate in another when LD patterns differ (Vilhjálmsen et al., 2015; Martin et al., 2019). As a result, PRS developed from

Eurocentric datasets shows steep performance declines when applied to African, Indigenous American, Asian, and admixed populations (Duncan et al., 2019; Wang et al., 2021). These population-genetic dynamics are a primary reason that PRS accuracy and calibration degrade as genetic distance from European reference samples increases (Martin et al., 2019; Wang et al., 2021).

Given that GWAS identifies “tag SNPs” that are statistically associated with causal variants via LD, these population-specific haplotype structures fundamentally shape GWAS results. A SNP that effectively tags a causal variant in European LD architecture may fail to do so in African or admixed genomes, where LD patterns differ substantially across populations (Vilhjálmsdóttir et al., 2015; International HapMap Consortium, 2007). Therefore, PRSs trained predominantly on Europeans depend on population-specific LD patterns and allele frequencies and will not reliably translate to other groups. This mismatch produces steep declines in PRS accuracy when applied to African, Indigenous American, Oceanian, and South Asian ancestry populations (Duncan et al., 2019; Martin et al., 2019). Multiple studies have shown that PRS accuracy declines as genetic distance from the European reference populations used in most GWAS training datasets increases (Martin et al., 2019; Wang et al., 2021).

This carries severe public health implications for these populations. This manner of miscalibration with polygenic risk scores disproportionately affects and misclassifies risk for a population that is already at risk and currently underserved by the healthcare system. On top of that, inaccurate predictions can distort screening eligibility, lead to inappropriate clinical decisions, and further entrench existing health disparities. As public health agencies consider bringing PRS into clinical screening and population-level prevention strategies, these inequities, if left unaddressed, represent significant, urgent scientific, ethical, and policy changes.

In this paper, the guiding question is: which equity metrics should public health agencies require before deploying PRS scores for screening, prevention, or population health? Addressing this question demands identifying which populations are underrepresented in major genomic datasets, understanding how ancestry and haplotype structure influence PRS performance, determining the sample size thresholds required for equitable predictive accuracy, and developing formal criteria for minimum evidence before deployment. This paper argues that equitable precision medicine is achievable only when genomic tools are built on data that meaningfully represent global human diversity. To support equitable PRS deployment, public

health agencies should require ancestry-specific performance metrics, enforce minimum representation thresholds in genomic datasets, mandate multi-ancestry validation, and adopt governance frameworks grounded in fairness, transparency, and reciprocity.

II. Scientific Background and Population Genetics Context

Genetic variation between people is structured in ways that influence the overall performance of GWAS and PRS models. While humans share the overwhelming majority of their DNA, there is still variation among global populations. Rosenberg et al. (2002) demonstrated that most genetic variation exists within populations rather than between “racial groups”. However, haplotypes exhibit substantial differences in frequency and structure across different ancestries.

The ethnographic population with the deepest history is the African population; this group exhibits shorter haplotype blocks and lower LD due to extensive recombination over generations (International HapMap Consortium, 2007; Tishkoff et al., 2009). In contrast, populations that migrated out of Africa experienced significant bottlenecks and showcase longer LD blocks with reduced diversity between haplotypes (Gravel et al., 2011; Schaffner et al., 2005). The importance here is that these differences are not merely academic; they directly determine how well SNPs can be identified as proxies for underlying traits in GWAS. Thus, when models are not representative of a specific population, it will be significantly more challenging to identify underlying patterns within that population.

Genome-wide association studies depend on detecting these statistical associations between SNPs and underlying disease traits. However, most variants detected in a GWAS are not themselves causal but rather tag SNPs linked to causal variants through linkage disequilibrium. Additionally, previous work has demonstrated that GWAS association signals are highly dependent on population-specific LD structures, which differ across global ancestries (Vilhjálmsdóttir et al., 2015; International HapMap Consortium, 2007). This means that a tag SNP identified as linked to Europeans may be in strong LD with a causal variant found in other Europeans but not at all in people of African ancestry. Consequently, genetic associations found in one population do not necessarily transfer to different populations.

Beyond that, the use of polygenic risk scores further propagates these same limitations. PRS effectively combines statistical evidence from GWAS to produce a single quantitative

genetic risk score, so the accuracy hinges on the validity of GWAS associations. Several studies showcase how poorly the transfer is between different population backgrounds. Duncan et al. (2019) observed that PRS derived from European datasets retain roughly complete accuracy in Europeans. Still, performance drops to around 60% in East Asian cohorts and to 40% or lower in African ancestry cohorts. Studies consistently show that PRS accuracy declines in proportion to genetic distance from the GWAS training population, with the steepest performance losses observed in African and Indigenous groups (Martin et al., 2019; Wang et al., 2021). Martin et al. (2019) warned that clinical deployment of Eurocentric PRS could exacerbate health disparities if applied without equity safeguards.

As such, the demographic composition of major genomic databases fundamentally reinforces these issues. While recent efforts, such as the All of Us Research Program and H3Africa, have significantly expanded non-European representation (All of Us Research Program Investigators, 2023), the overall statistical architecture of polygenic risk scores remains dominated by its Eurocentric origins.

III. Global Underrepresentation and Public Health Consequences

The significant underrepresentation of diverse populations in genomic datasets is a persistent systemic issue that has extraordinary implications for the future of health equity. Sirugo et al. (2019) estimated that more than three-quarters of all GWAS participants are of European ancestry, even though Europeans are a minority globally. Whilst people of African ancestry are represented in less than 2% of GWAS samples, other minority groups, such as Indigenous American, Oceanian, Middle Eastern, and South Asian groups, are also severely underrepresented.

This underrepresentation extends across disease domains. In psychiatric, cardiometabolic, autoimmune, and cancer-related GWAS, all are overwhelmingly of European ancestry (Sirugo et al. 2019). This, of course, results in effect size estimates through polygenic risk scores that poorly reflect global genetic variation. The structural consequence is therefore clear: tag SNPs may fail to track causal variants in populations with different LD patterns. These scientific limitations translate directly into public health inequities. PRS miscalibration leads to misclassification in risk stratification, inappropriate screening recommendations, and unequal access to preventive interventions. For example, PRS for conditions such as breast cancer or

cardiovascular disease may accurately stratify European ancestry patients while systematically underestimating or overestimating risk in African or Indigenous populations. Martin et al. (2019) caution that such disparities may exacerbate existing inequities in chronic disease outcomes. Pharmacogenomic predictions face similar problems, as drug metabolism variants differ substantially across ancestries (Duello et al., 2021).

As such, these inequities raise significant legal, social, and ethical concerns. Race is frequently and primarily improperly used as a pseudo proxy for genetic ancestry in genomic research, which both obscures biological distinctions and reinforces social stereotypes (Duello et al., 2021). Historically, many of these communities have faced significant patterns of exclusion and exploitation by the general population and the scientific community, which contribute to extreme distrust of such systems. Bioethics scholars emphasize that biobank governance must incorporate principles of reciprocity, transparency, and community partnership to ensure that populations contributing DNA also receive equitable benefits (HHS/NIH, 2023). Without such safeguards, genomic medicine risks perpetuating forms of genetic colonialism, whereas technological advantages accrue disproportionately to already privileged populations, whilst leaving struggling populations without the same tools.

IV. Mapping Missingness and Determining Sample Size Requirements

The underlying question at the heart of this issue is how to identify precisely who is missing from existing genomic databases and what level of representation is needed to accurately define a polygenic risk score. This involves analyzing ancestry proportions in large datasets, examining LD structure, and establishing sample size requirements informed by demographic and evolutionary history. Looking at larger genomics databases (GWAS Catalog, UK Biobank, TOPMed, H3Africa, etc.), we can see persistent disparities in ancestral representation (Sirugo et al., 2019; Bycroft et al., 2018). Recent pan-ancestry analyses of the UK Biobank continue to reveal substantial power imbalances and ancestry-linked differences in effect sizes across global populations (Karczewski et al., 2025). While the All of Us database has expanded significantly in terms of non-European participation, decades of research biased toward Europeans mean that most disease-associated variants and genetic tools are primarily optimized for European populations. This, in turn, leads to non-European groups producing lower precision in both the ability to detect SNPs of any significance and the magnitude of their importance.

Groups of African ancestry, as previously stated, exhibit shorter haplotype blocks and reduced LD relative to non-African populations, providing an especially vivid example of these challenges (Tishkoff et al., 2009; International HapMap Consortium, 2007). As a result of these differences, attempting to identify associations in populations of African ancestry requires larger sample sizes of three to four times as many participants to achieve the same statistical power attained in European datasets (Vilhjálmsdóttir et al., 2015; Wang et al., 2021). As such, these dynamics have very direct implications for the validity and accuracy of PRS. Ensuring equitable PRS performance, therefore, may require targeted recruitment efforts such as ancestry-stratified sample size thresholds.

Creating a method for mapping what we are missing from the datasets merely provides a foundation for addressing the inequity that we are currently dealing with. By identifying gaps at both the population and disease levels, we can develop more strategic plans to create databases that are more inclusive for GWAS frameworks. Without this groundwork, however, any public health application proposing to utilize PRS is likely only to reinforce existing disparities rather than mitigate them.

V. Proposed Equity Metrics and Policy Recommendations for PRS

Deployment

To ensure equitable adoption of polygenic risk scores, we need to adopt explicit, ancestry-aware performance standards grounded in mathematical principles from machine learning fairness. At a minimum, we should hold public health agencies to requiring an ancestry-stratified model evaluation that specifically examines outcomes for ancestral populations. This could include metrics such as explained variance (R^2) and AUROC computed separately for each significant ancestral population. In addition, models should report PRS transferability ratios, which effectively show the potential decline in accuracy when models are applied to an ancestral group for which they were not trained. This is a fundamental indicator that there is a structural inequity in the underlying data and that they would need to revisit their dataset.

The Fairness in Machine Learning survey by Caton and Haas (2024) argues that quantitative fairness rests on three foundational criteria: Independence, Separation, and Sufficiency. While the independence criterion mathematically requires that predictions be

entirely unrelated to group membership, this is neither realistic nor appropriate in the case of genomics, since genetic architectures, allele frequencies, and LD differ substantially across global populations. However, we can utilize the other two criteria, the Separation criteria, which focus on Equalized Odds and Equal Opportunity, and require that the true-positive and false-positive rates be comparable between groups. As such, in the case of implementing this for PRS, we aim to align these rates across ancestries, ensuring that specific groups are not exposed to systematically higher false-negative rates in disease screening. Sufficiency is tied to calibration fairness, meaning that individuals with the same predicted risk should have the same actual probability of disease regardless of their ancestral background. This tenet may be significantly more difficult to uphold, as aforementioned disease-associated SNPs and LD structure differ quite substantially across ancestral populations.

For policy implementation, these fairness metrics offer concrete tools for regulatory oversight. Agencies should require ancestry-stratified reporting of PRS accuracy, error rates, and calibration, aligning with the Separation and Sufficiency conditions. They must also mandate multi-ancestry validation and the disclosure of performance disparities, recognizing the survey's recommendation that fairness must be measured using multiple metrics since no single criterion is universally achievable. Furthermore, representation thresholds are essential, as populations with greater haplotype diversity require larger sample sizes, and insufficient representation makes fairness criteria mathematically unattainable. Embedding these fairness principles into PRS evaluation provides a mathematically rigorous and ethically aligned framework for equitable genomic medicine.

VI. Future Directions: Multi-Teacher Distillation Modeling Frameworks for Equitable PRS

To create a more equitable system for identifying polygenic risk scores, it may be beneficial to lean more on ML principles in the future. Primarily, as we can obtain more diverse genomic datasets that truly represent a genetic population, we could utilize a multi-teacher distillation architecture. Effectively, we would have several ancestry-specific models that act as “domain experts.” In this framework, each teacher model is trained exclusively on a single ancestral group, allowing it to capture population-specific patterns in LD, allele frequencies, and haplotype structures that a model trained on a global population couldn't. These ancestry-specific

teacher models then serve as specialists, and the predictions would be representative only of their population.

From there, we could use these teacher models trained on specific ancestral populations to integrate and distill information into a “student” model. Essentially, this setup allows us to create a student model that should learn a global multi-ancestry representation of genetic risk scores, informed by the strengths of each specialized teacher model. More importantly, we can calibrate fairness constraints grounded in Separation and Sufficiency during the distillation of this model to ensure equalized error rates across ancestries. This framework has significant potential and could be particularly valuable in the future. However, it's necessary to recognize that obtaining sufficient active participation in dataset creation is essential before jumping into model development. As it stands, without adequate data, we may be able to build an ML model to predict PRS; however, it would likely still uphold hegemonic norms until we have more diversity in the underlying data.

VII. Conclusion

Despite advances in the field, significant constraints remain on the development and equitable deployment of polygenic risk scores. A large number of ancestral groups remain significantly underrepresented in genomic databases, limiting statistical power and hindering our ability to estimate effect sizes. In addition to this, labeling only by ancestry may also oversimplify the highly complex patterns of human diversity that have occurred as a result of migration and colonization, which may obscure within-population heterogeneity, especially among admixed individuals (Rosenberg et al., 2002; Sirugo et al., 2019; 1000 Genomes Project Consortium, 2015). Lastly, fairness-aware approaches first require the construction of diverse datasets and ethical oversight, which are simply not yet the standard in genomic infrastructure.

However, it is both possible and achievable to build an equitable genomic future with deliberate design. If we want to improve the accuracy of PRS across ancestries substantially, we can address representation gaps, enforce metrics, and, in the future, possibly implement multi-teacher modeling frameworks. To better validate results, public health agencies should require ancestry-stratified validation and representation thresholds before fully integrating PRS into screening or prevention programs. The reality is that precision medicine can only be called such and truly be precise when it accurately reflects global human diversity. By implementing

these tools into PRS, we can create tools that empower all populations rather than simply reinforce current health disparities.

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